iii. Impeller

Impeller is a rotating part of mechanism made up of cast iron or steel metal in centrifugal pump as shown in figure 4.25. By centrifugal force the water is delivered with uniform pressure without any vibration.



Fig. 4.25 **impeller**

Types

1. Open type impeller.
2. Closed type impeller.

iv. Shaft

It is made of stainless steel, to avoid rusts in the shaft while using salty water. The portion of the shaft which works inside the casing is usually fitted with gun-metal sleeve and hence no chance of depreciation will occur. The gun metal sleeve can be replaced when it gets worn out, and it increases the life span of the shaft.

v. Rope and box

The main purpose of the rope and box is,

i. to prevent leakage of air on the suction side.
ii. to prevent leakage of water on the delivery side due to pressure.

The packing material consists of rings of soft cotton, woven yarn, impregnated with graphite and tallow. The gland bolts should only be tightened lightly, to prevent leakage.

Modern pumps are fitted with mechanical leak-proof seals as shown in figure 4.26. Basically it is made up of softer materials like rubber, leather or plastic with nice finishing. It keeps cool with the water inside the pump. Otherwise it causes friction in the shaft and gets heated. This makes the pump function to get stop. It is very important that a centrifugal pump should not be allowed to run without water.

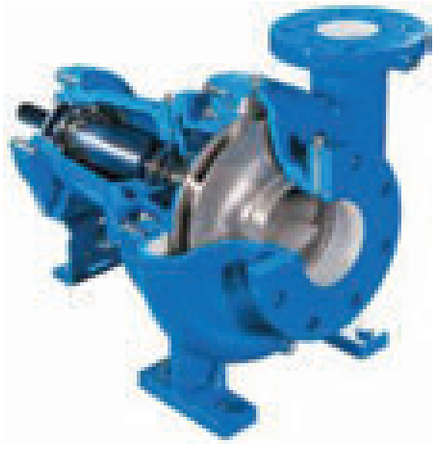


Fig. 4.26 **Internal system**

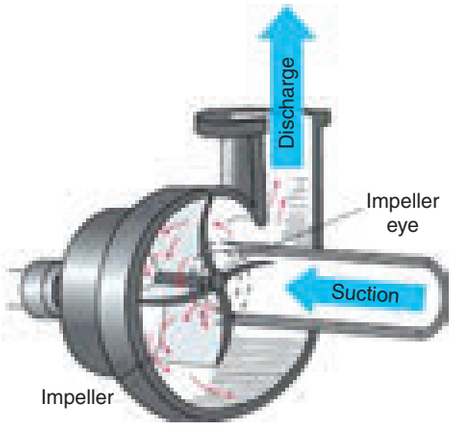


Fig. 4.27 **Internal cutting look**

vi. Bearing

Ball, roller and bush bearings are often used. Usually ball and roller bearings are lubricated with oil and grease.

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